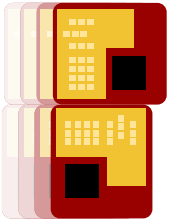




Towards Practical Bi-Static Polarimetric Imaging Using Commodity mmWave Radars for Material Sensing



Xinghua Sun, Akshay Gadre

University of Washington

Material Information is a Fundamental Need



Non-contact Security Check



Quality Control



Liquid Testing

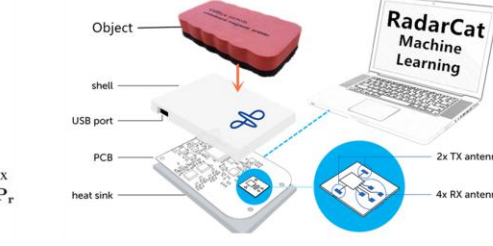
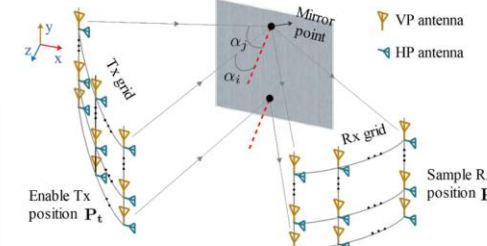
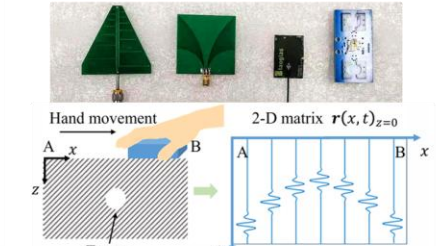
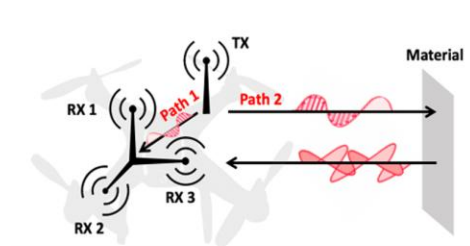
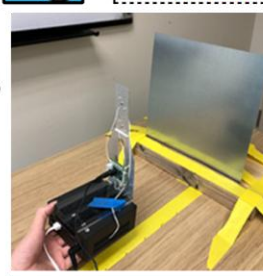
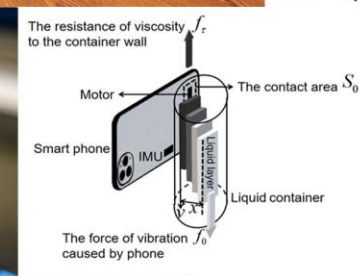
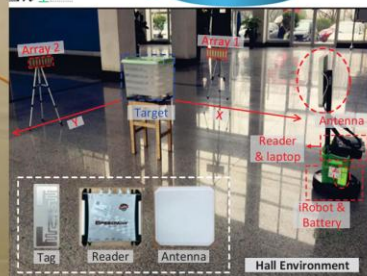
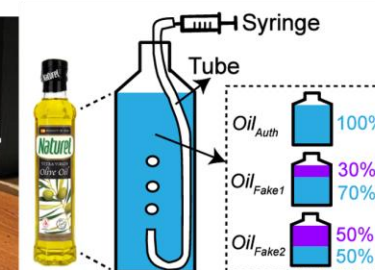
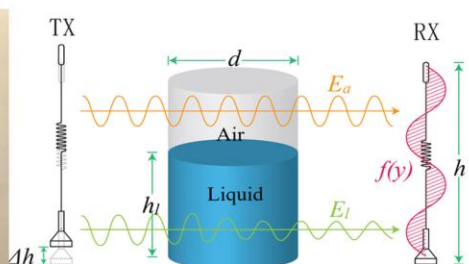
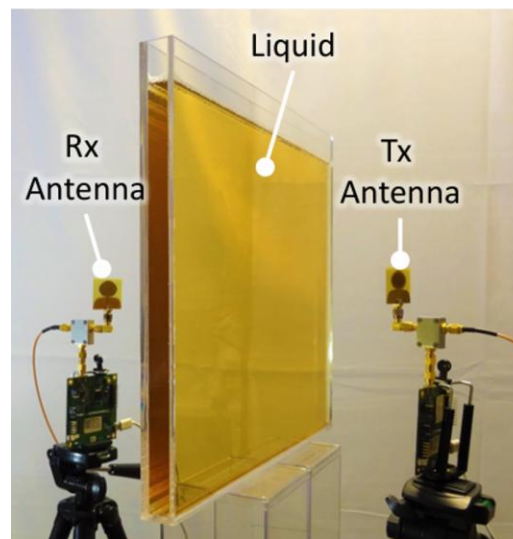


Autonomous Robotics

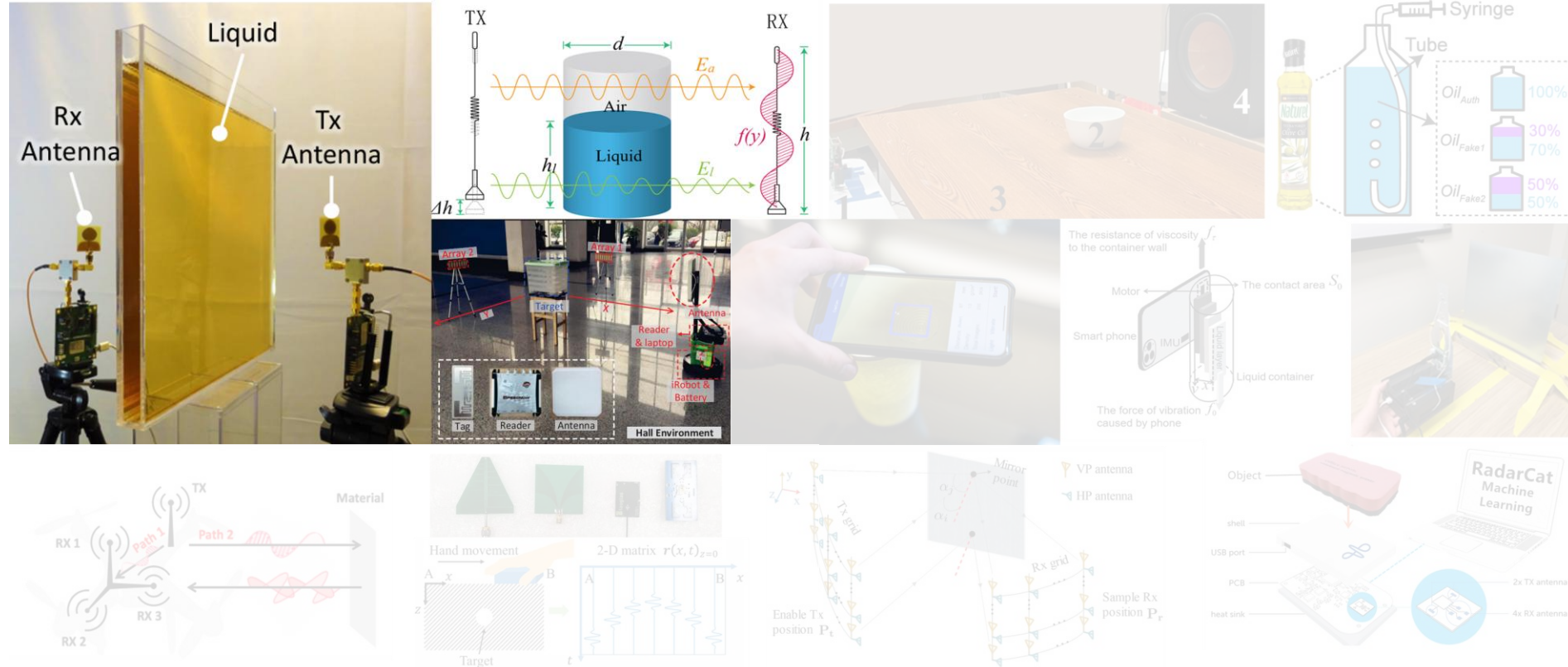


Autonomous Driving

Prior Art



Prior Art

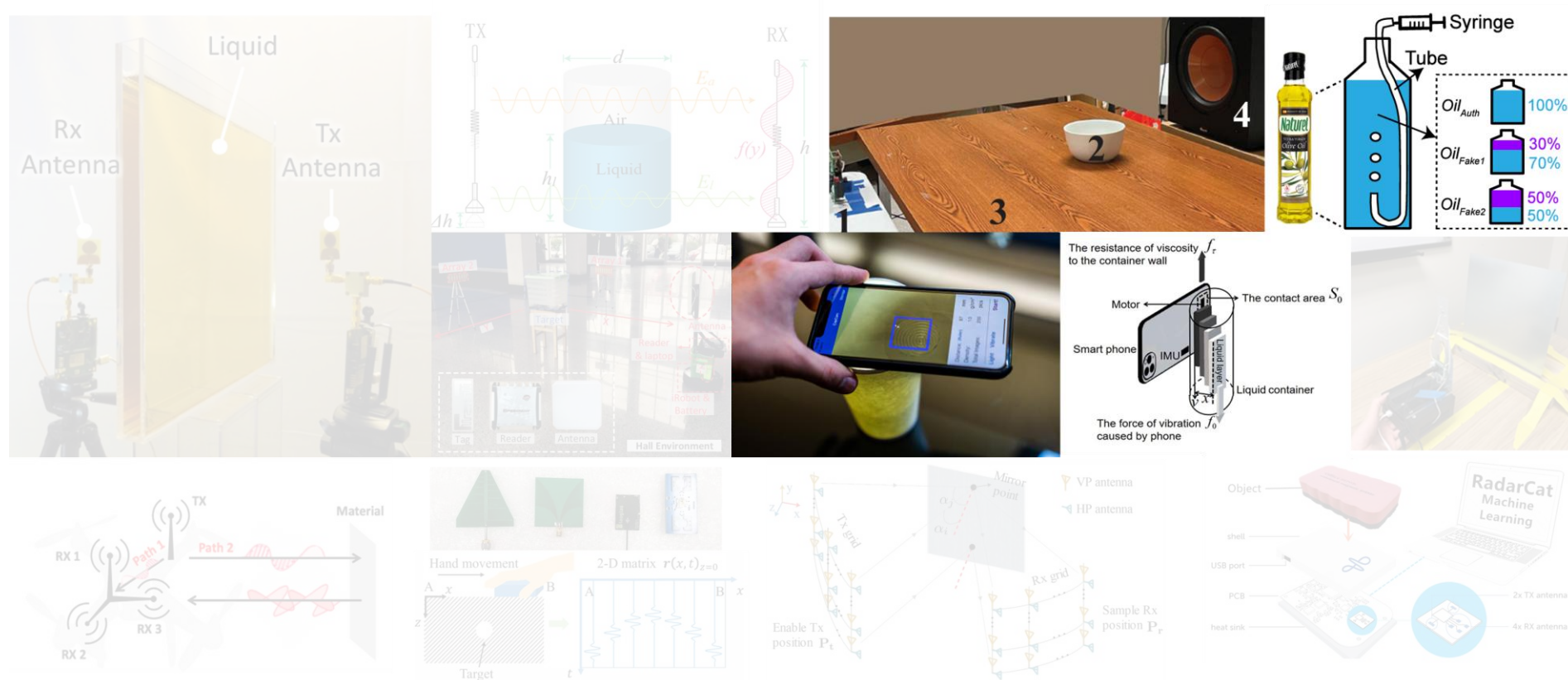


LiquidD: A Wireless Liquid Identifier, MobiSys '18

TagScan: Simultaneous Target Imaging and Material Identification with Commodity RFID Devices, MobiCom '17

LiqRay: non-invasive and fine-grained liquid recognition system, MobiCom '22

Prior Art



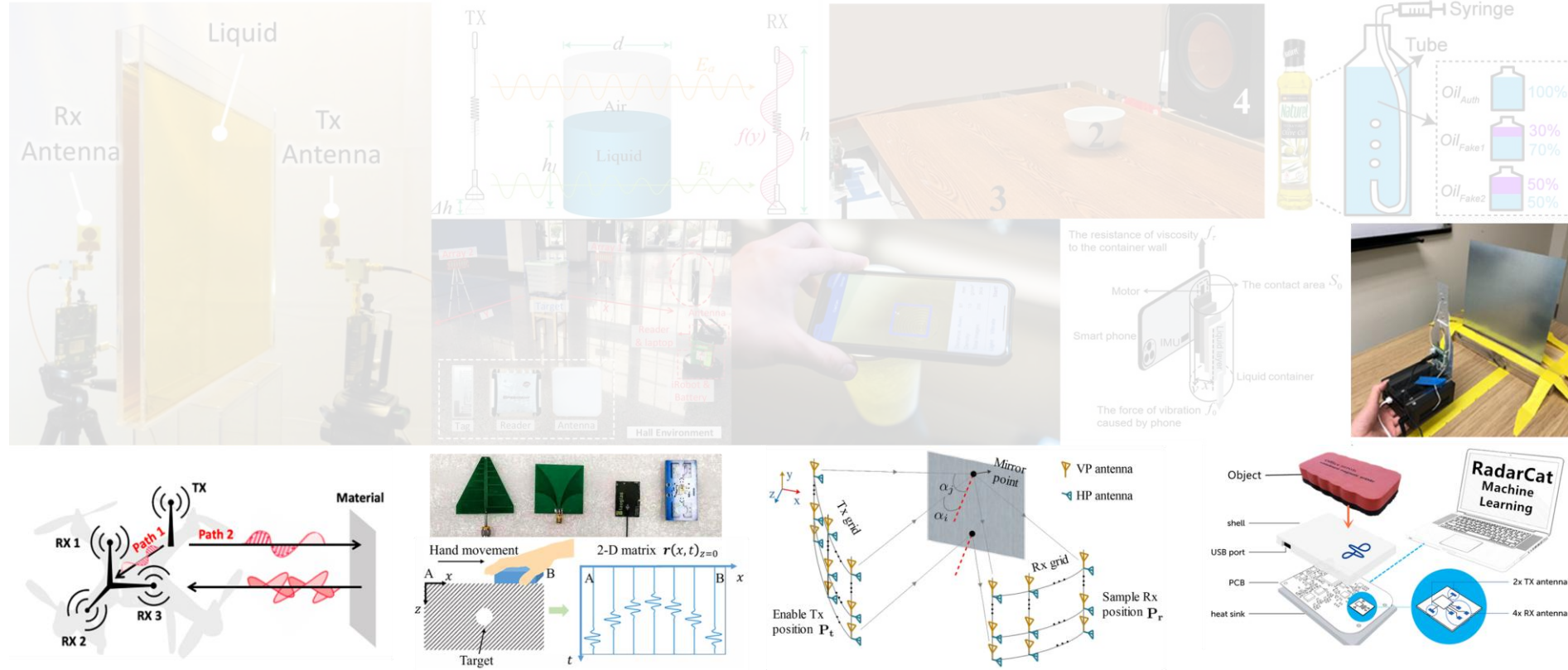
Liquid Testing with Your Smartphone, MobiSys '19

Vi-liquid: unknown liquid identification with your smartphone vibration, MobiCom '21

Detecting counterfeit liquid food products in a sealed bottle using a smartphone camera, MobiSys '22

Contactless material identification with millimeter wave vibrometry, MobiCom '23

Prior Art



RadarCat: Radar Categorization for Input & Interaction, UIST '16
 On the Feasibility of Wi-Fi Based Material Sensing, MobiCom '19
 mSense: Towards Mobile Material Sensing with a Single Millimeter Wave Radio. Ubicomp '20
 SiWa: see into walls via deep UWB radar, MobiCom '21
 Wi-Painter: Fine-grained Material Identification and Image Delineation Using COTS WiFi Devices, Ubicomp '23

Prior Art

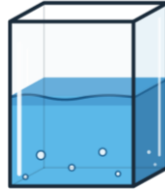
	Penetration-based	Mechanical-based	Backscatter-based
Accuracy	Accurate	Accurate	Inaccurate
Practicality	Only work for liquids	Invasive Object-centric	Different material forms Less-limitation to objects

Trade-off: Practicality vs. Accuracy

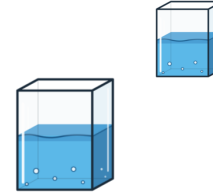
Factors Affecting the Signal



Shape



Size



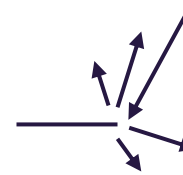
Location



Orientation



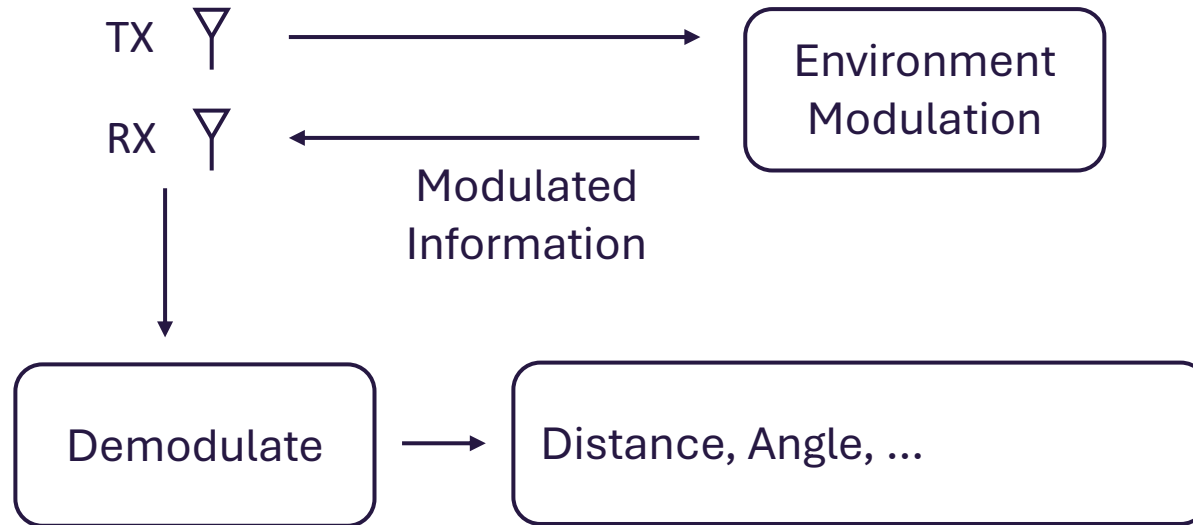
Roughness



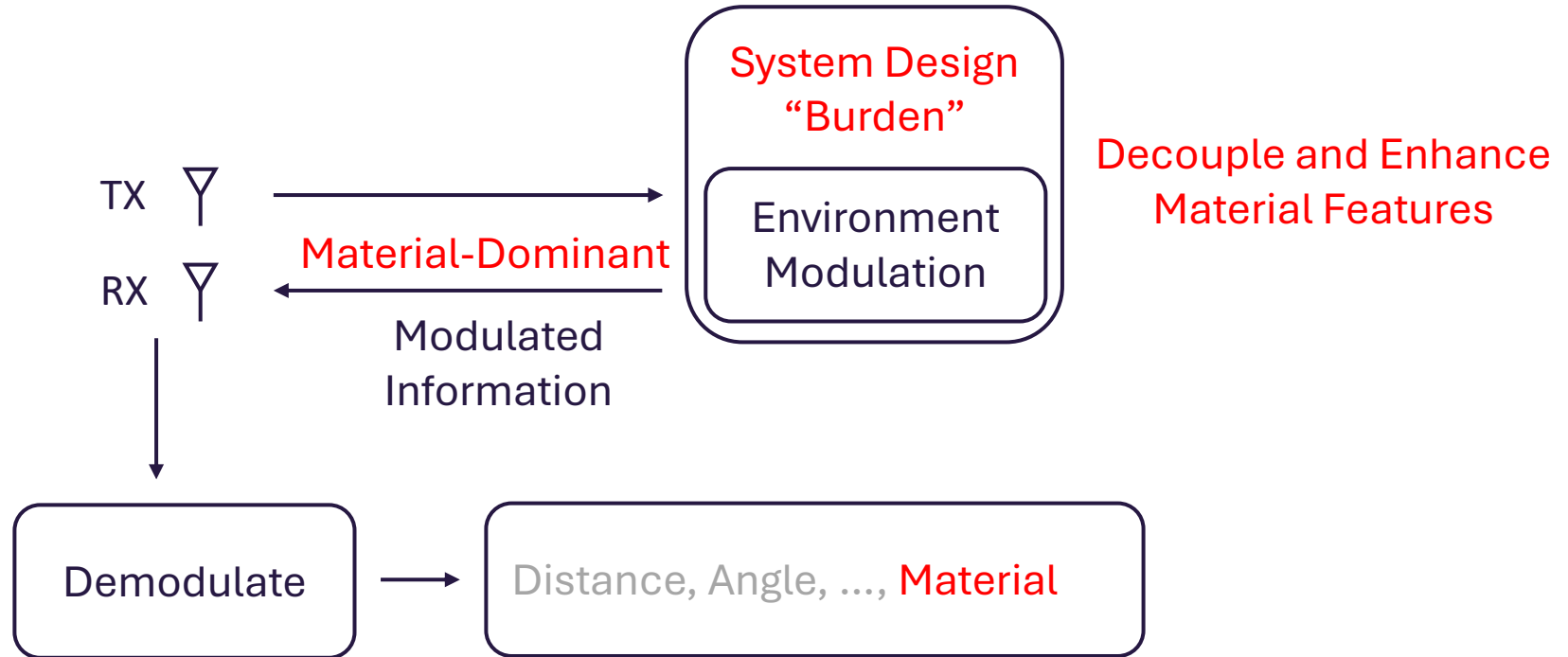
Edge Diffraction

Material isn't the dominant feature

Sensing is Non-cooperative Communication

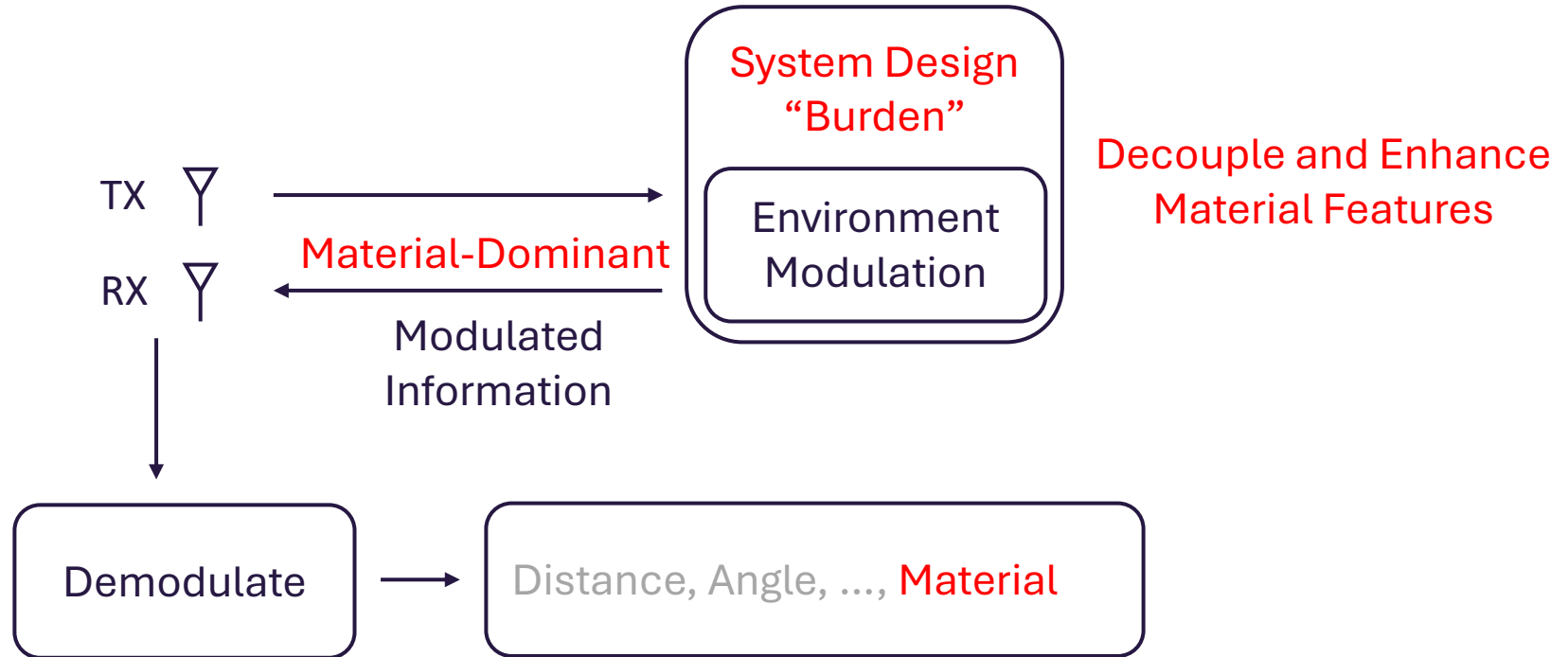


Sensing is Non-cooperative Communication

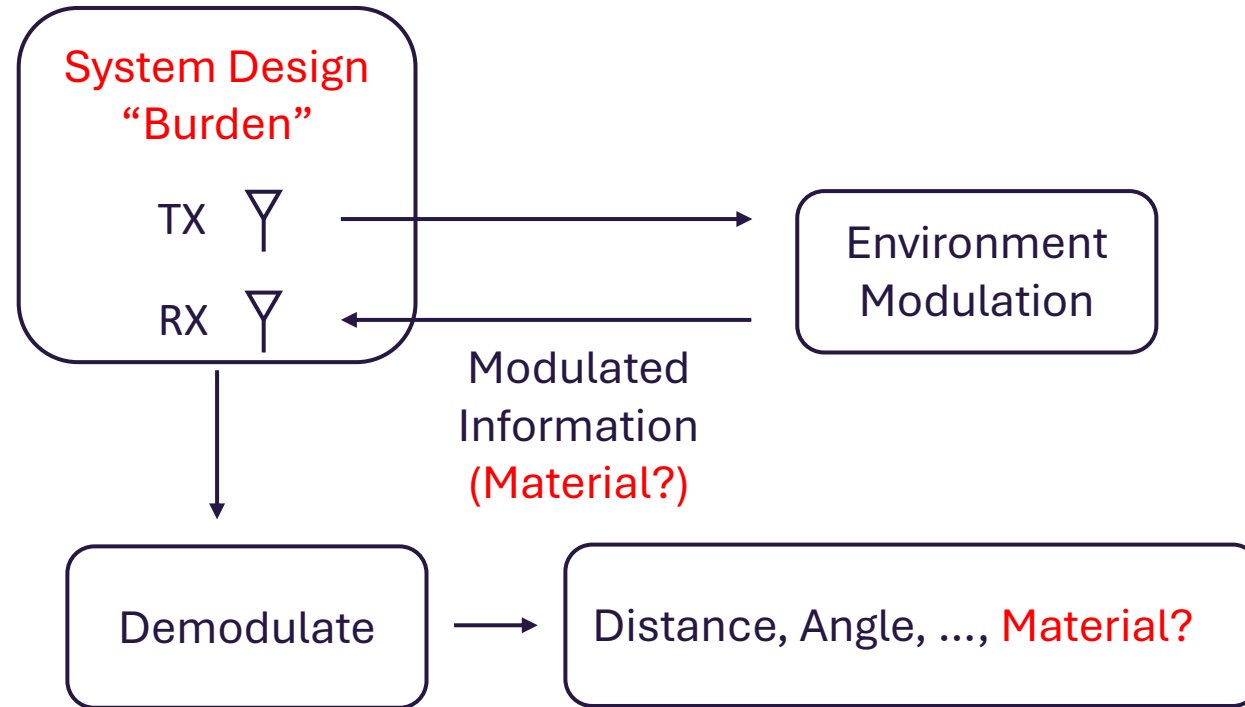


Can we identify materials in **practical** environments **accurately**
and non-invasively?

Sensing is Non-cooperative Communication

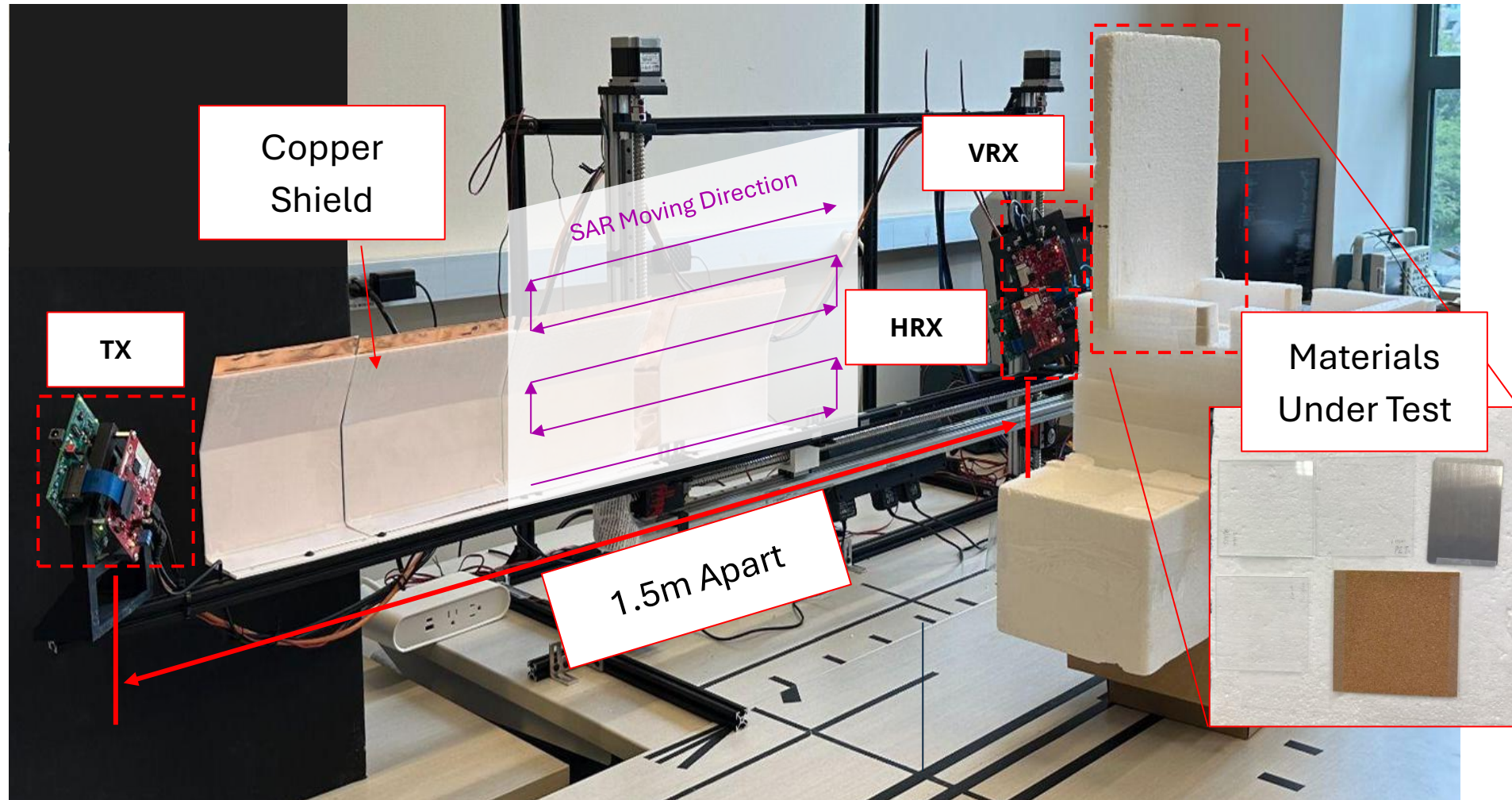


Sensing is Non-cooperative Communication

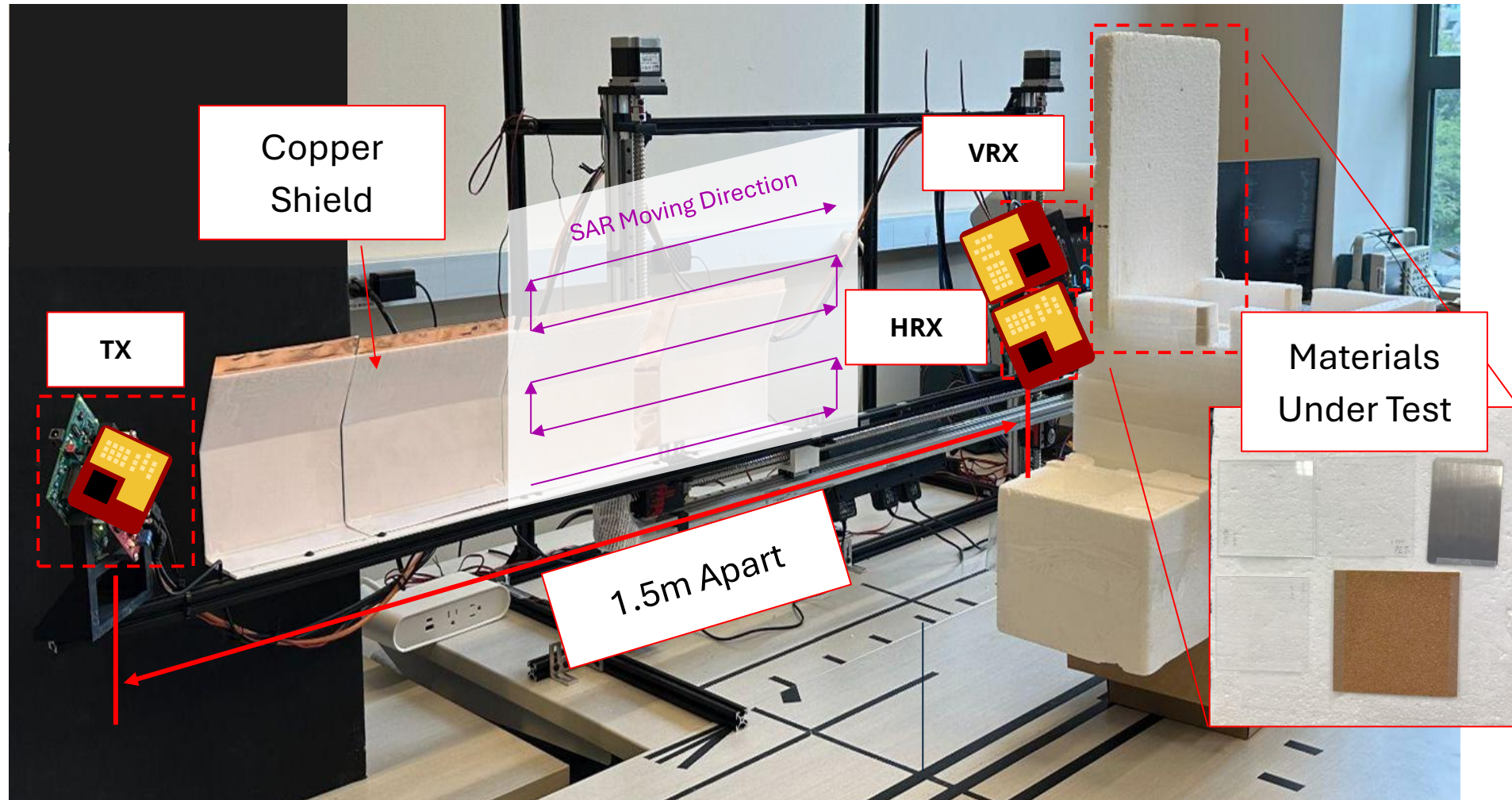


Move the burden from objects to sensors

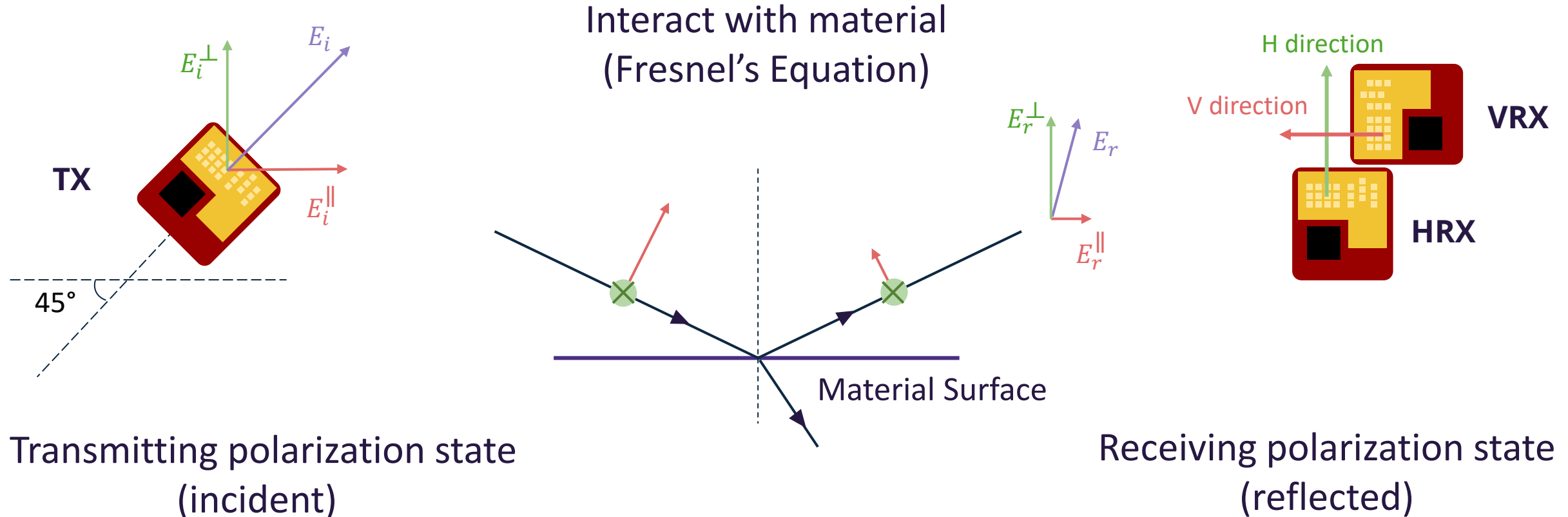
POLySight: Bi-static Polarimetric SAR Imaging System



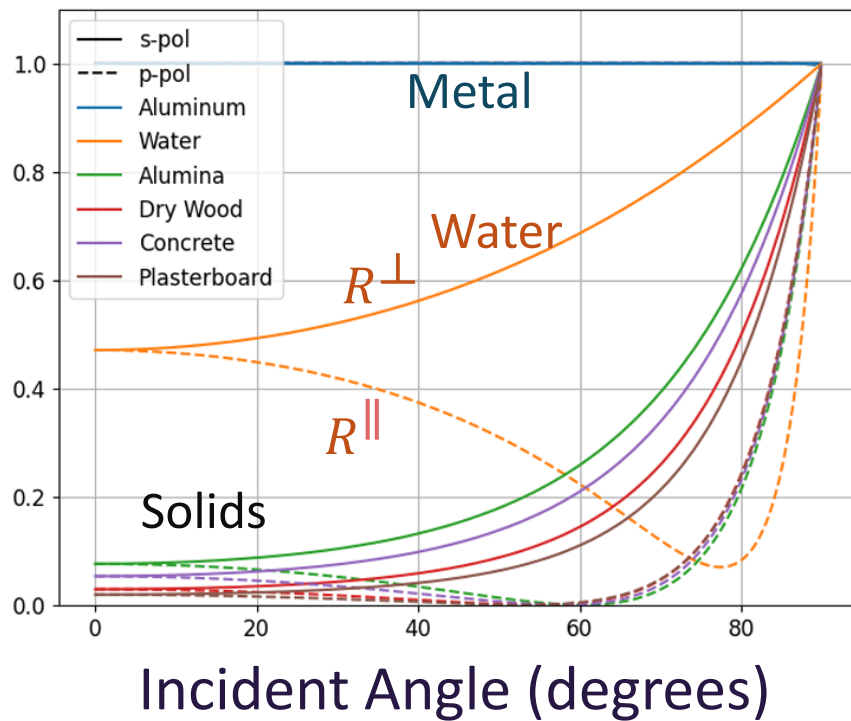
POLySight: Bi-static Polarimetric SAR Imaging System



New Dimension: Polarization

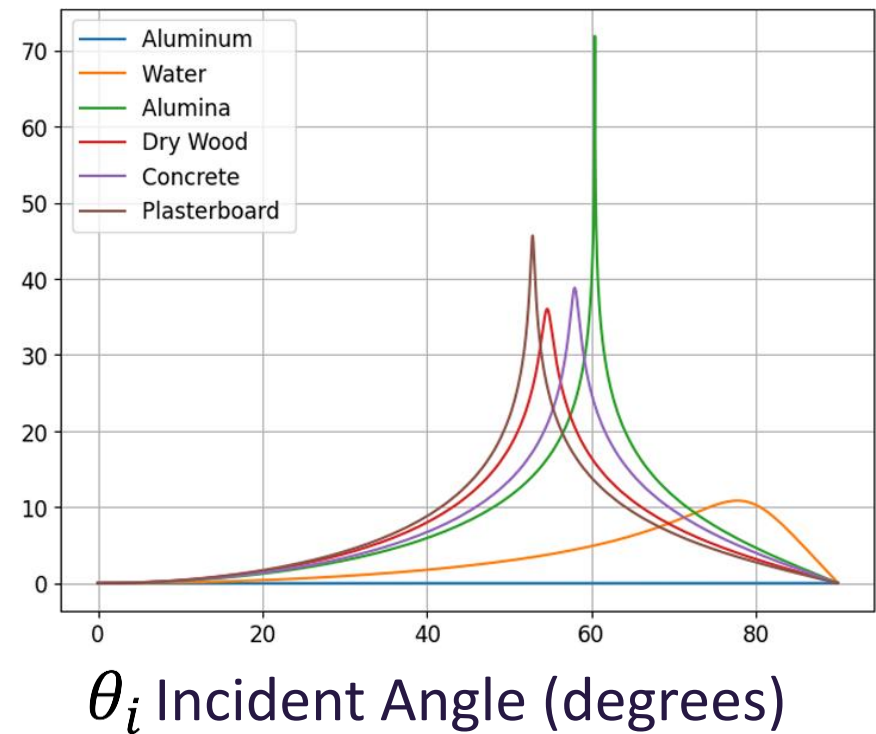


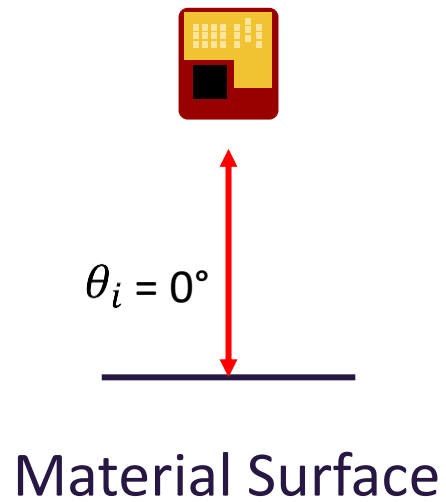
Reflectance $R^\perp$ and $R^\parallel$



Polarization Ratio

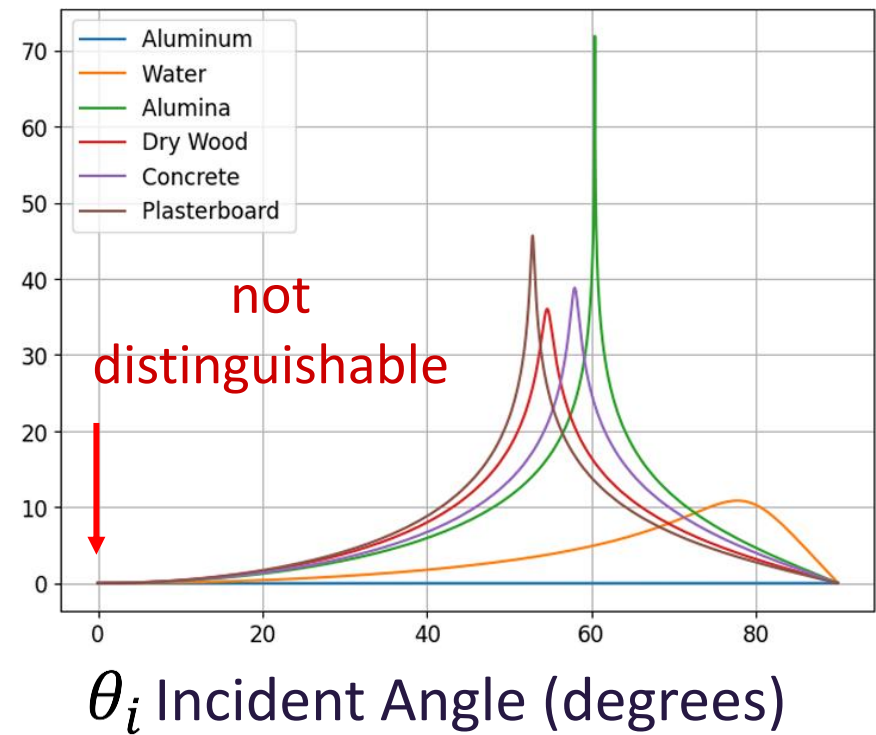
$$|\mathcal{P}| = \left| \frac{R^\perp}{R^\parallel} \right| \text{ (dB)}$$

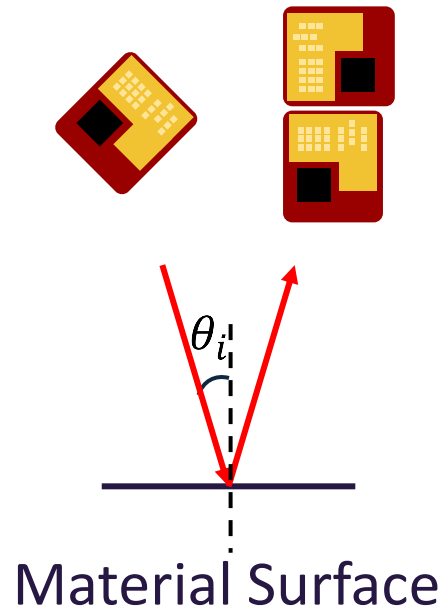




Polarization Ratio

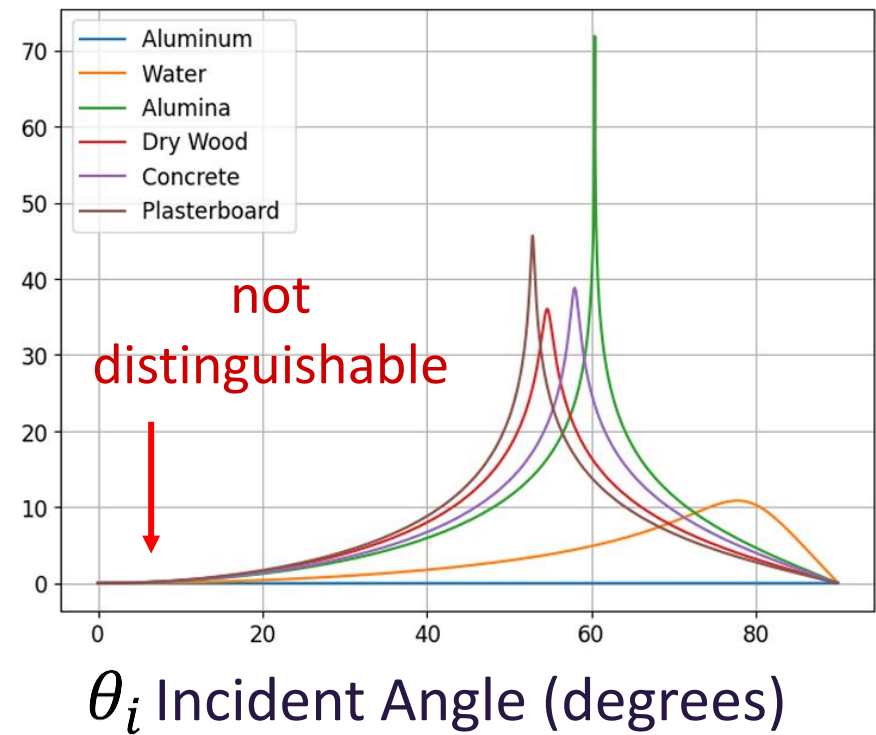
$$|\mathcal{P}| = \left| \frac{R^\perp}{R^\parallel} \right| \text{ (dB)}$$

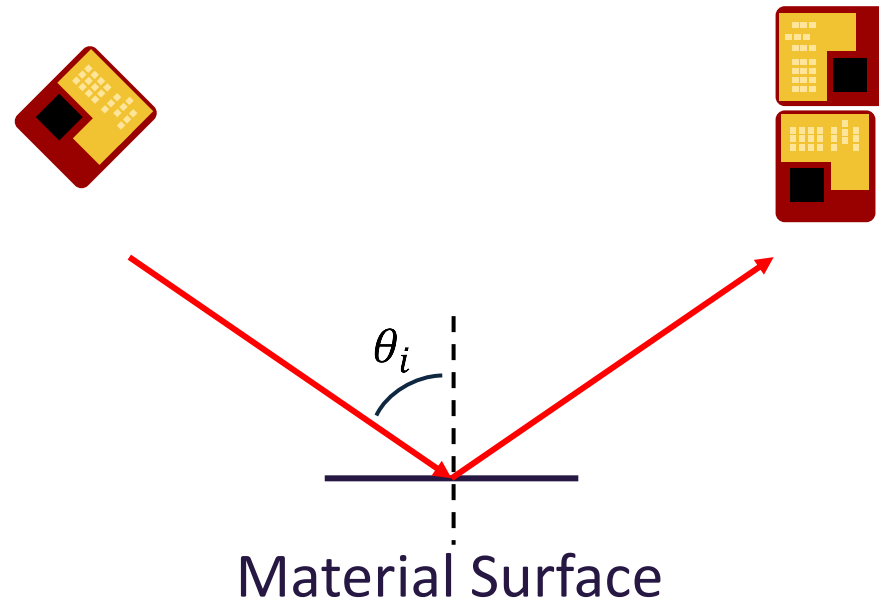




Polarization Ratio

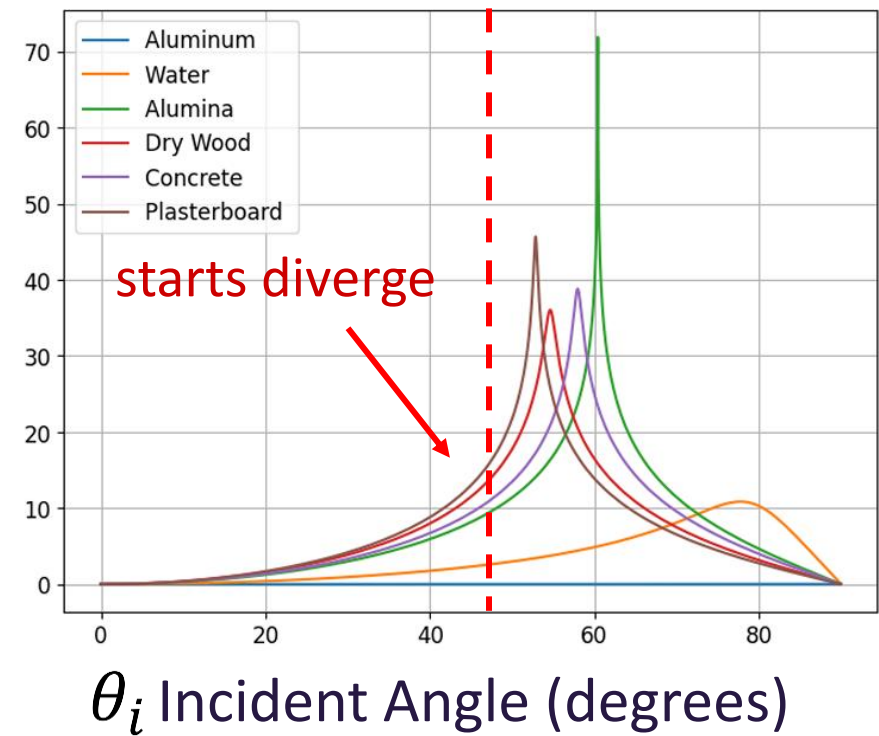
$$|\mathcal{P}| = \left| \frac{R^\perp}{R^\parallel} \right| \text{ (dB)}$$



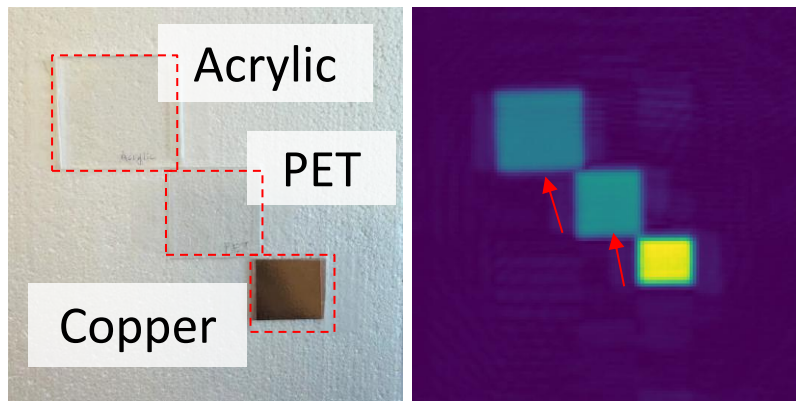


Polarization Ratio

$$|\mathcal{P}| = \left| \frac{R^\perp}{R^\parallel} \right| \text{ (dB)}$$

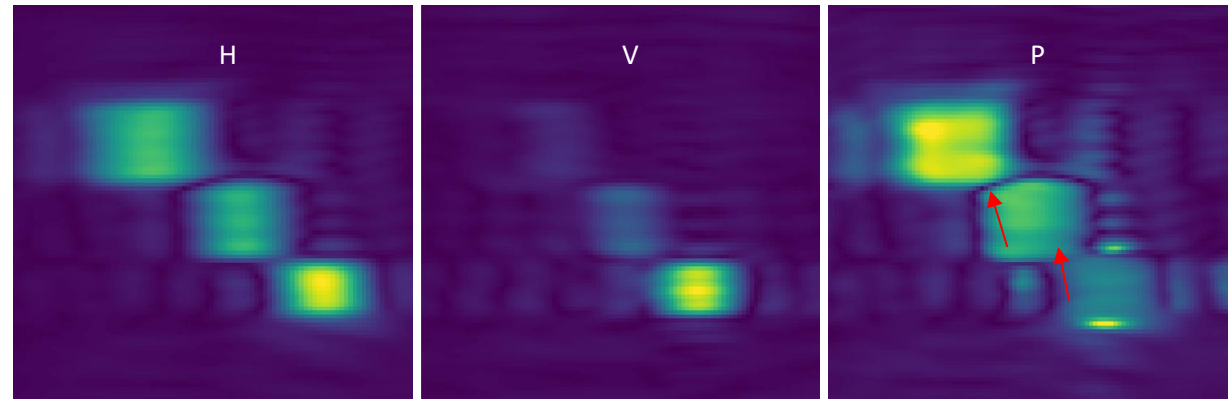


Conventional SAR

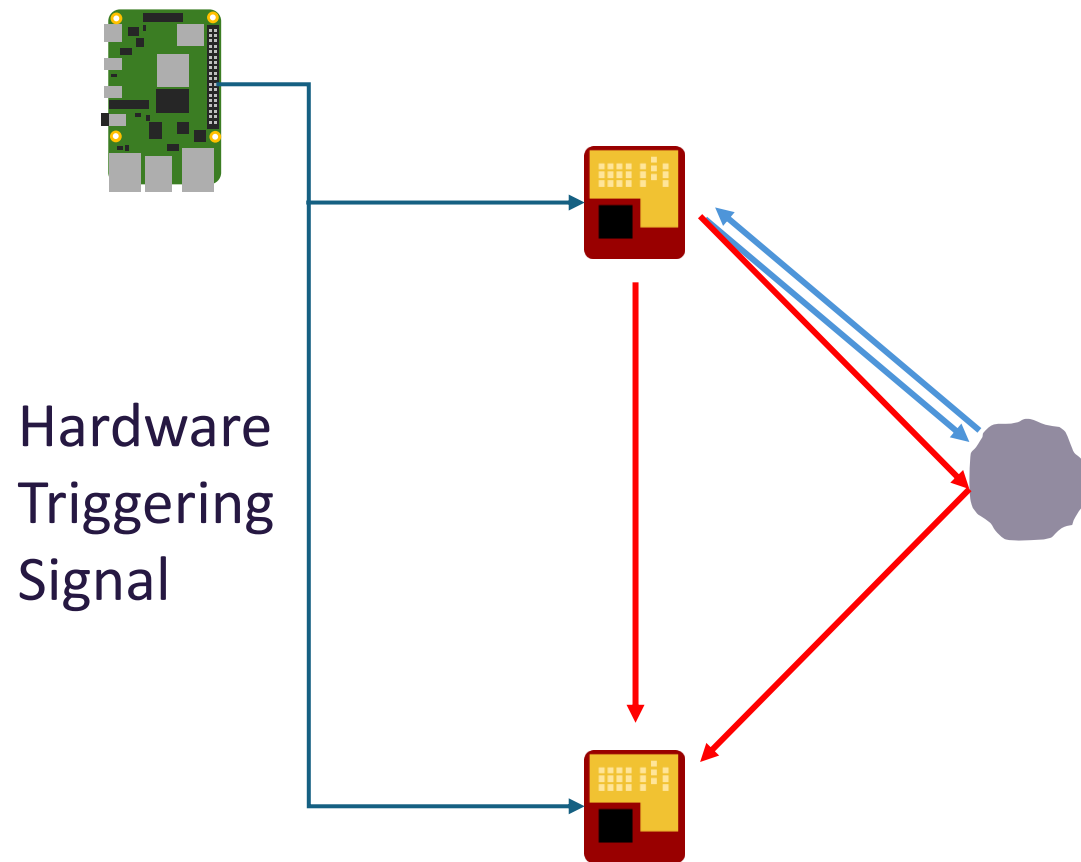


Not Distinguishable

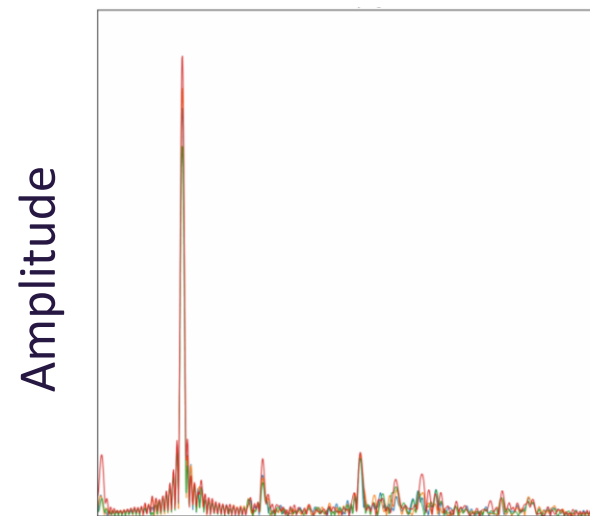
PolySight



Clearly Distinguishable

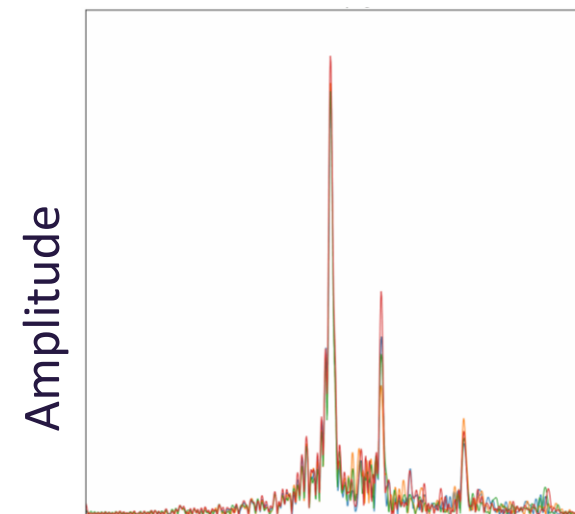


Mono-static Channel

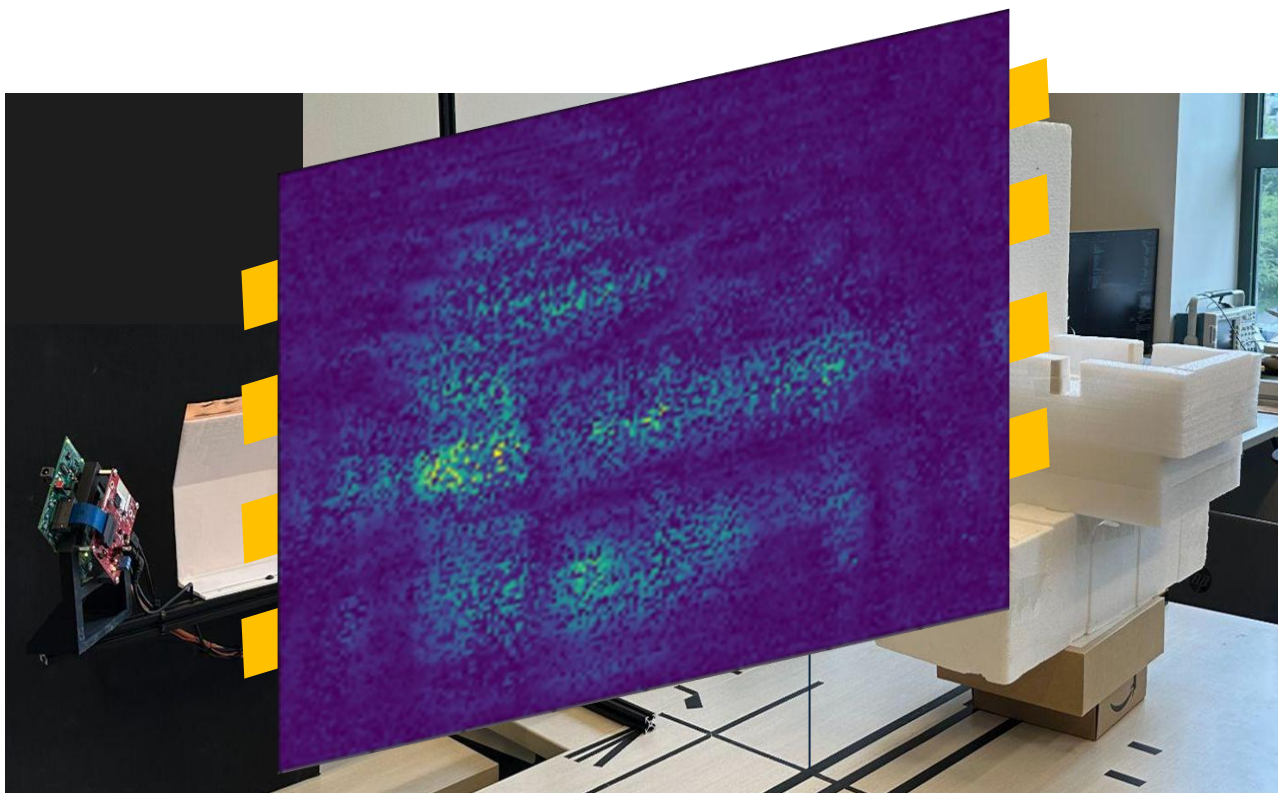


Range Profile

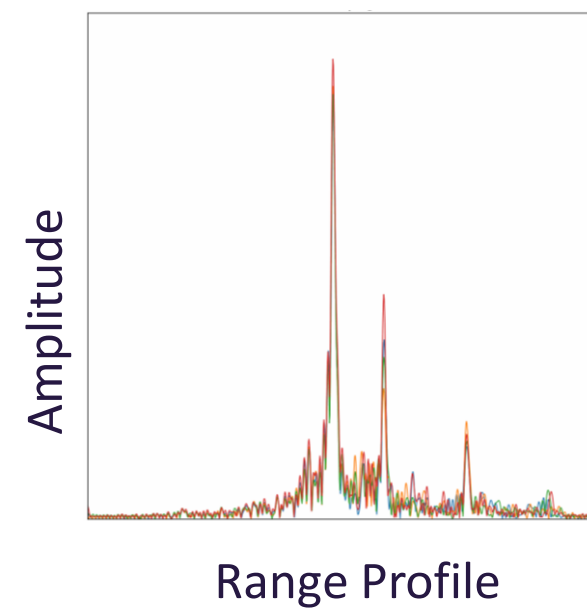
Bi-static Channel

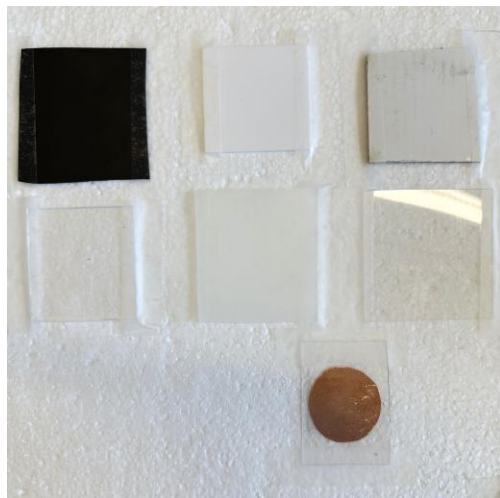


Range Profile

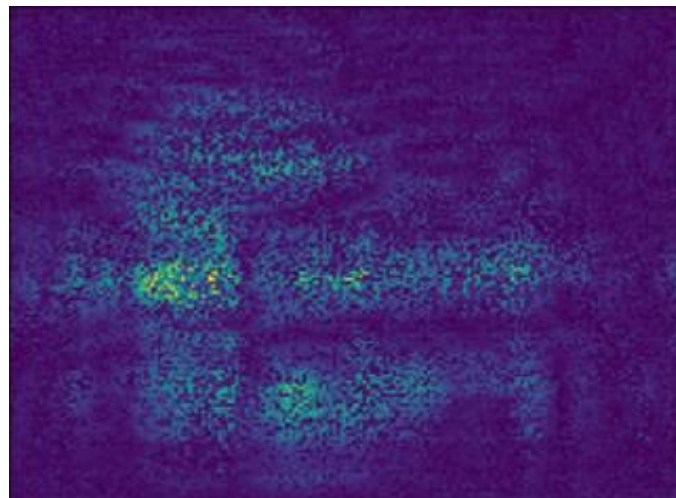


Unsynchronized
Bi-static Channel

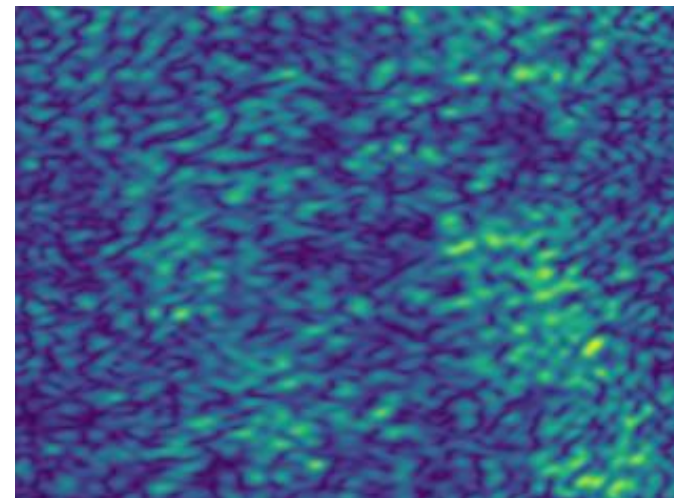




Imaged Materials

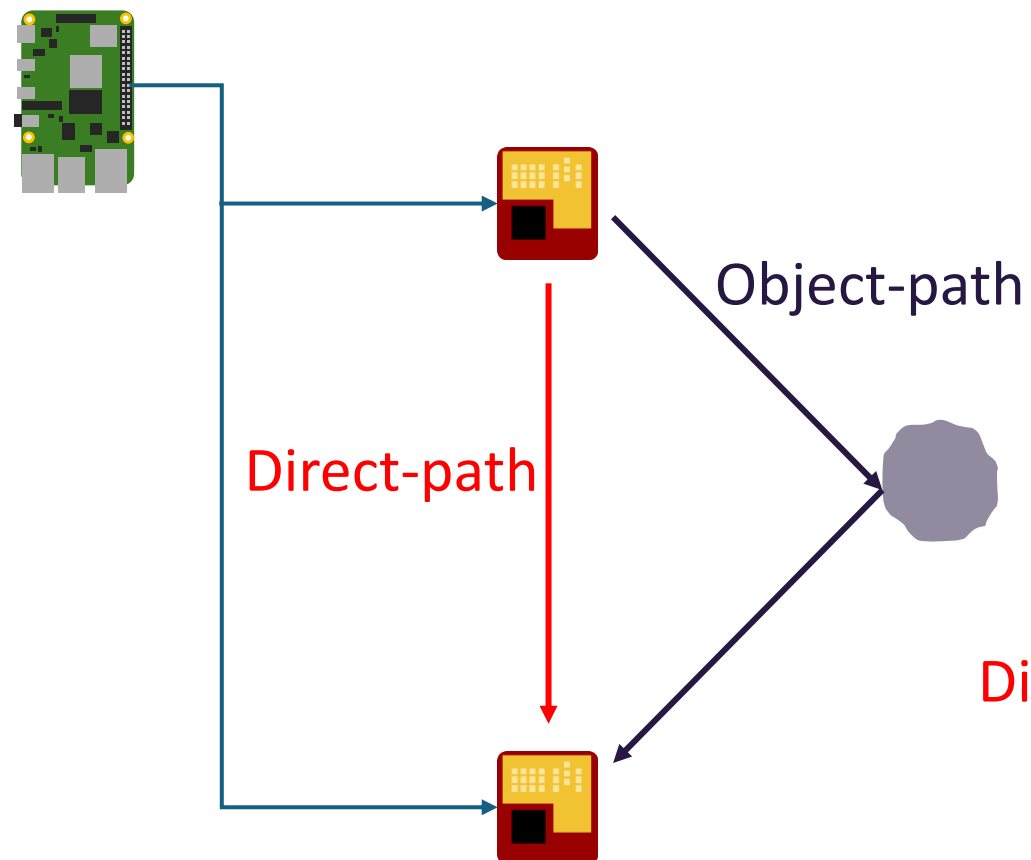


Unsynchronized
Bi-static Radar Cube

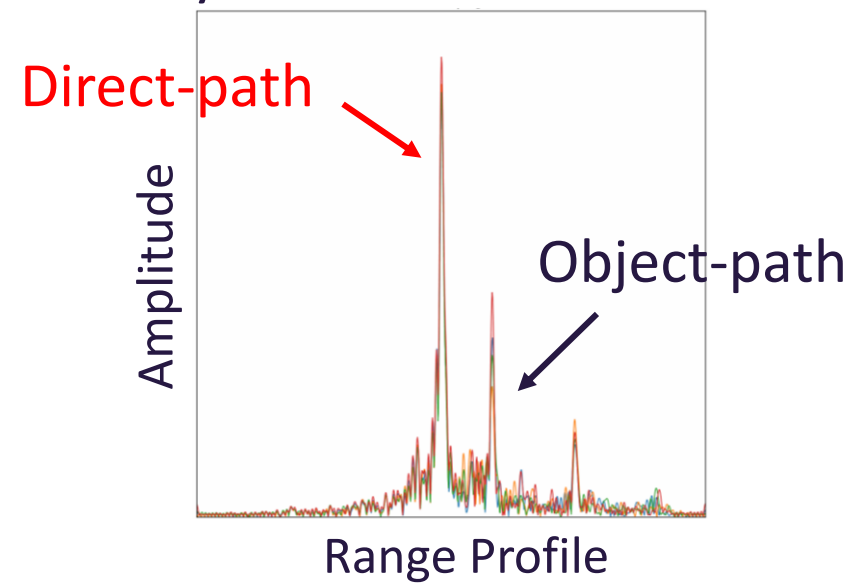


Unsynchronized
Bi-static SAR Image

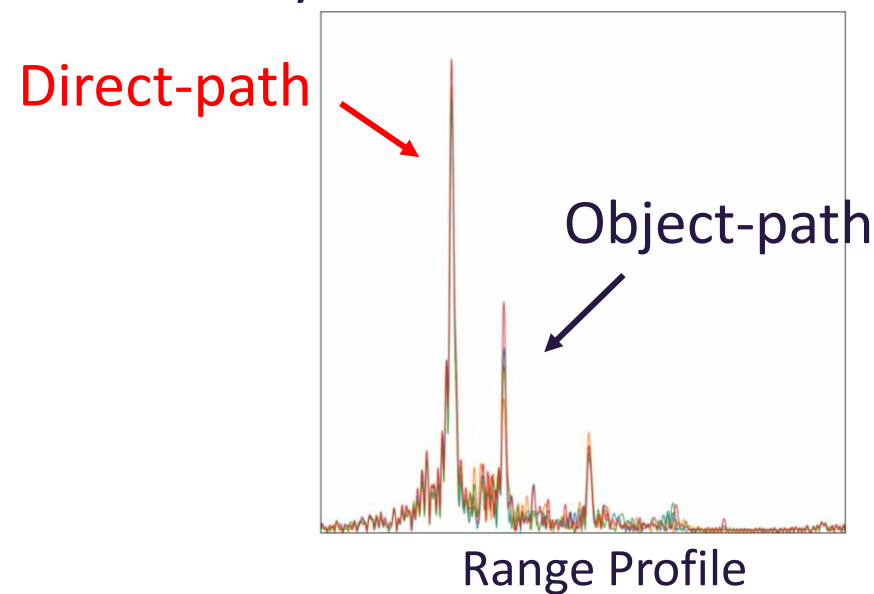
Frequency and phase offsets needs to be calibrated

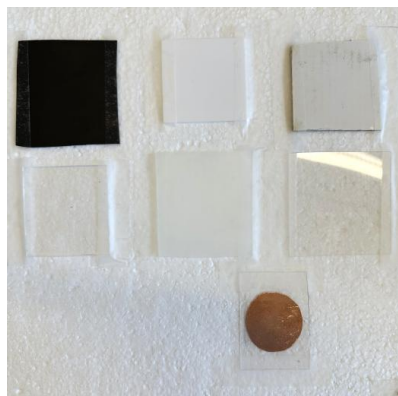


Unsynced Bi-static Channel

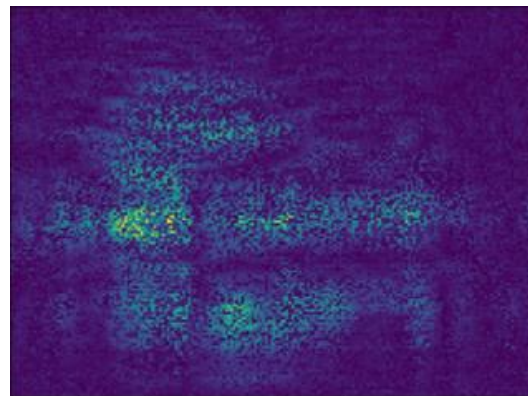


Synced Bi-static Channel

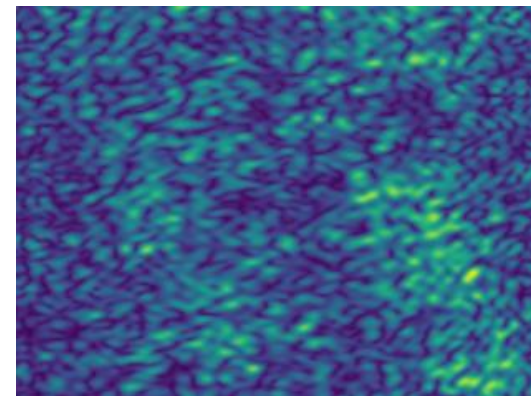




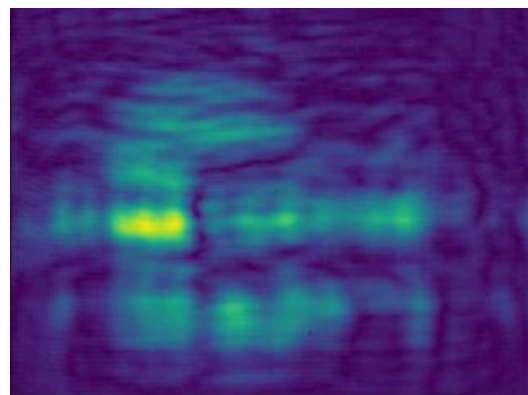
Imaged Materials



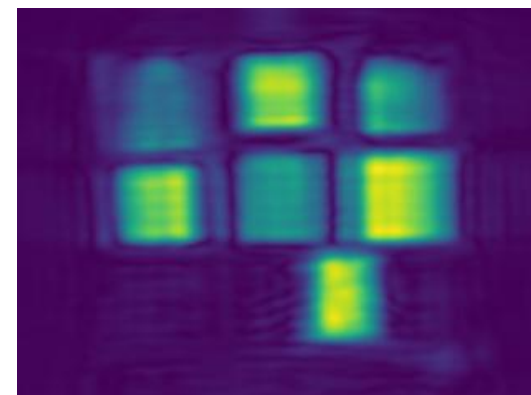
Unsynchronized
Bi-static Radar Cube



Unsynchronized
Bi-static SAR Image

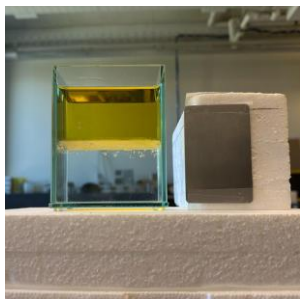


Synchronized
Bi-static Radar Cube

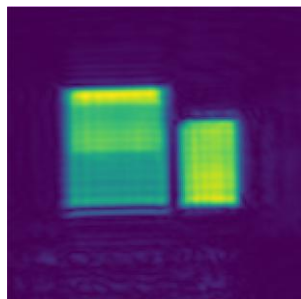


Synchronized
Bi-static SAR Image

Materials



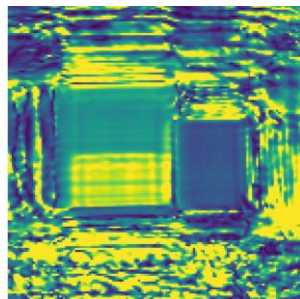
H



V



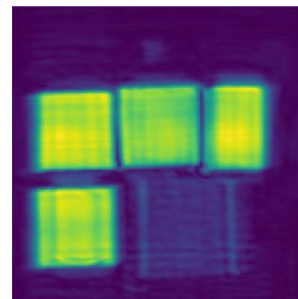
P



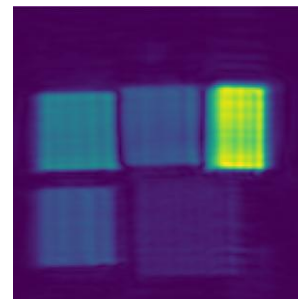
Materials



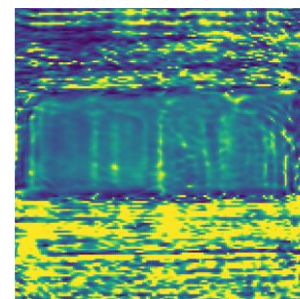
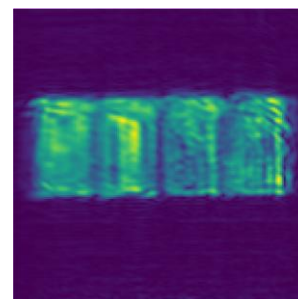
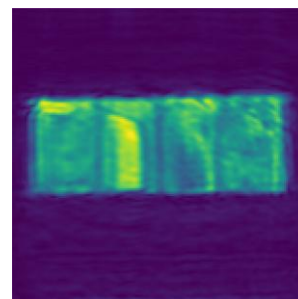
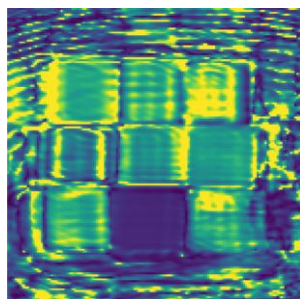
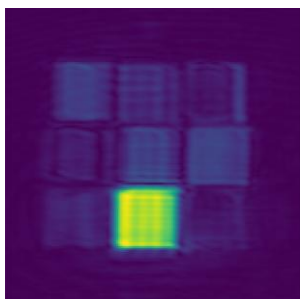
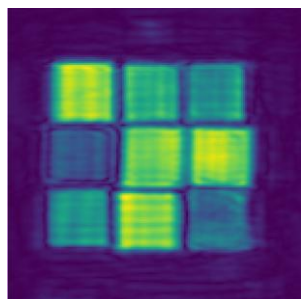
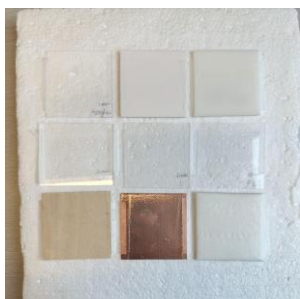
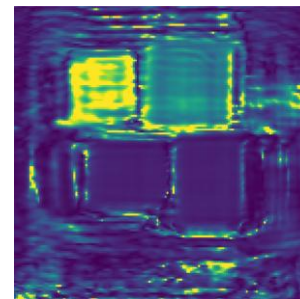
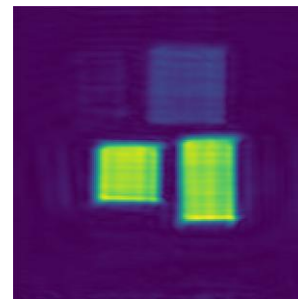
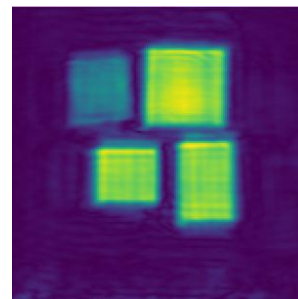
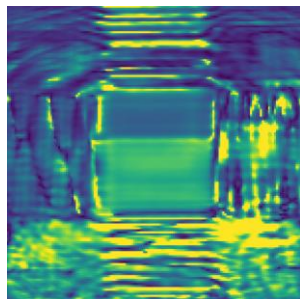
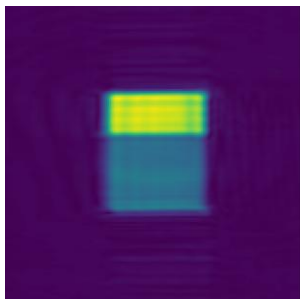
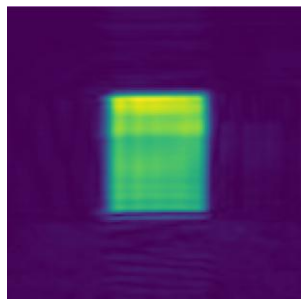
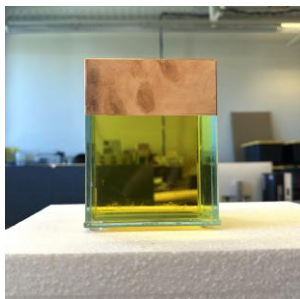
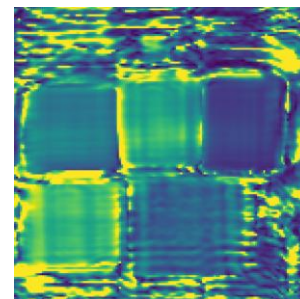
H



V



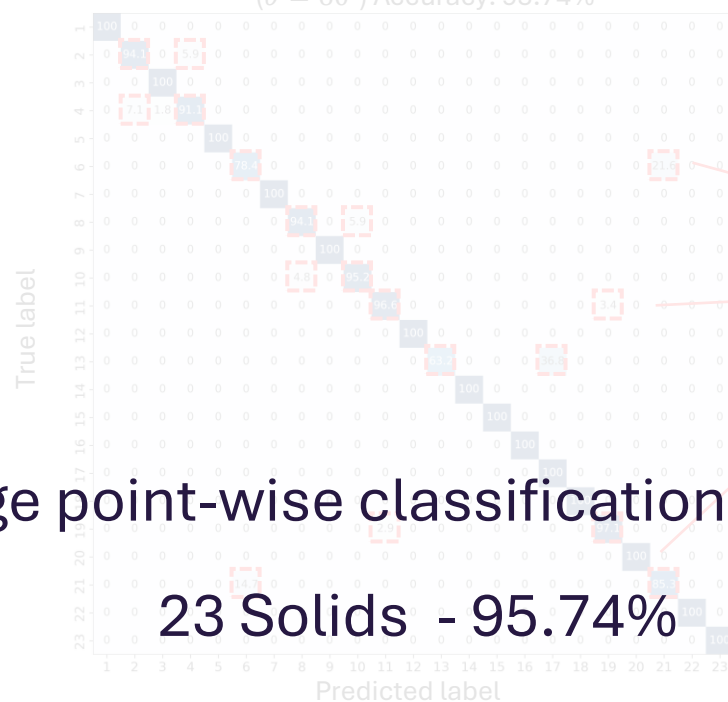
P



w/o/ PolySight

Solids using H channel only
($\theta = 60^\circ$) Accuracy: 48.46%

w/ PolySight

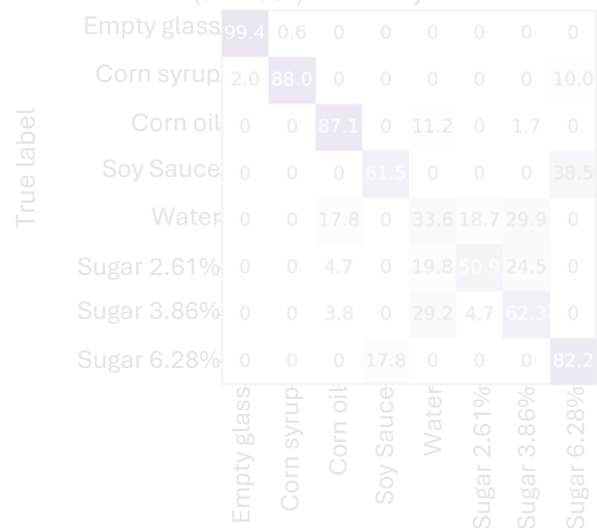
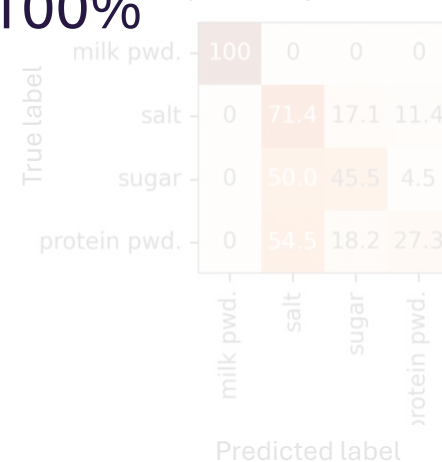
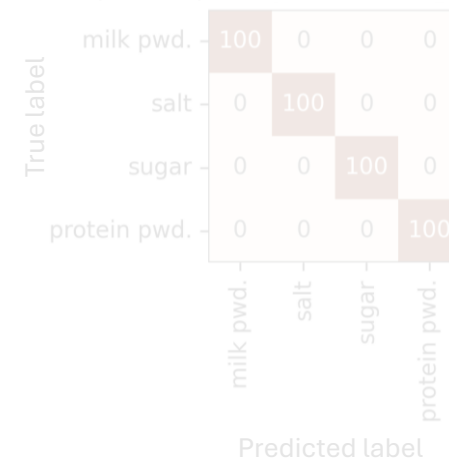
Solids using H, V, P channels
($\theta = 60^\circ$) Accuracy: 95.74%Fine-grained Classification
Solids at multiple incident angles
($\theta = 60^\circ, 55^\circ$) Accuracy: 99.34%

Average point-wise classification accuracy:

23 Solids - 95.74%

8 Liquids - 96.50%

4 Powders - 100%

Liquids using H channel only
($\theta = 55^\circ$) Accuracy: 72.87%Liquids using H, V, P channels
($\theta = 55^\circ$) Accuracy: 96.50%Powders using H channel only
($\theta = 55^\circ$) Accuracy: 58.69%Powders using H, V, P channels
($\theta = 55^\circ$) Accuracy: 100%

w/o/ PolySight

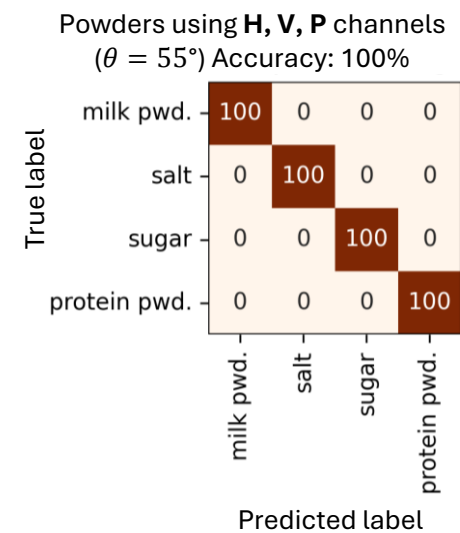
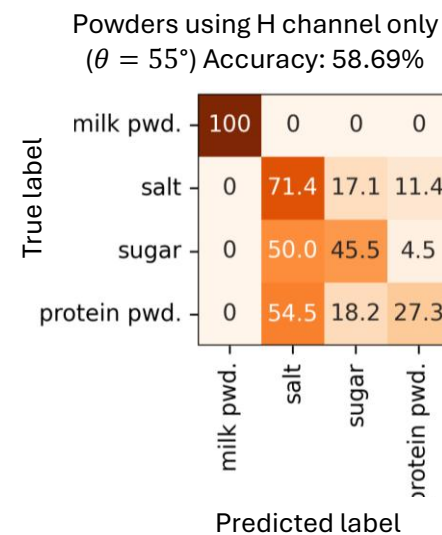
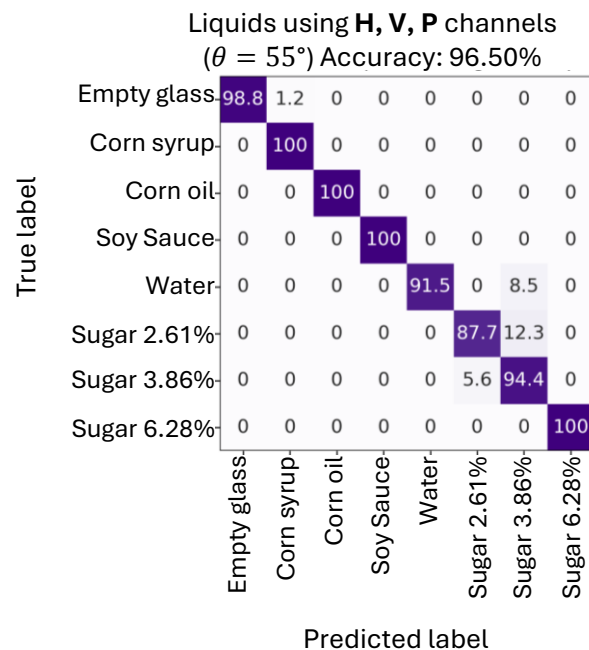
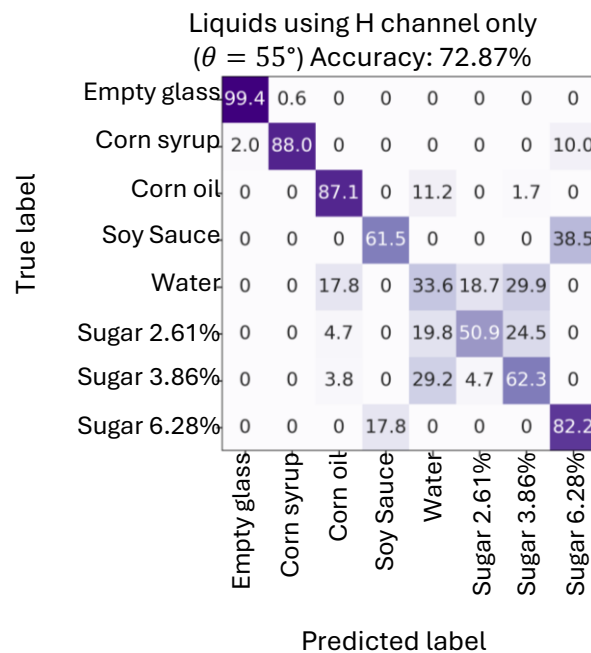
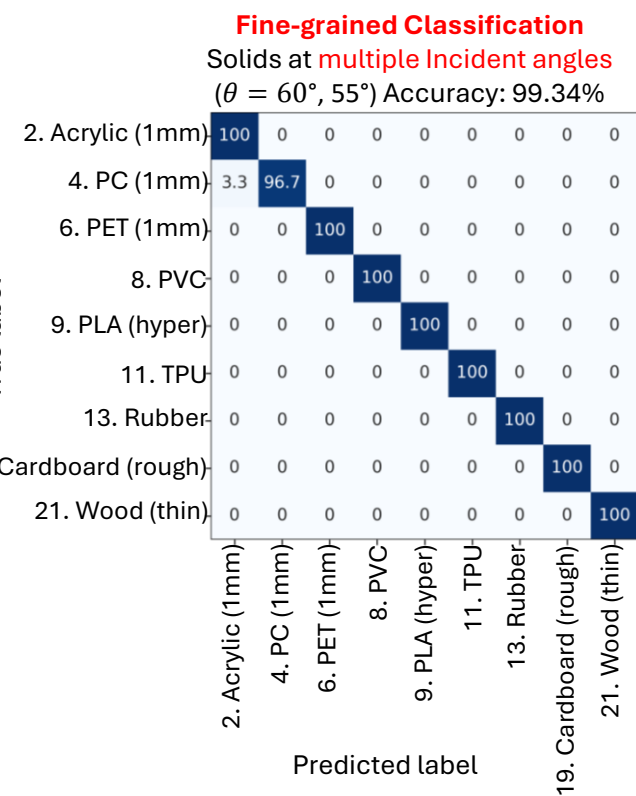
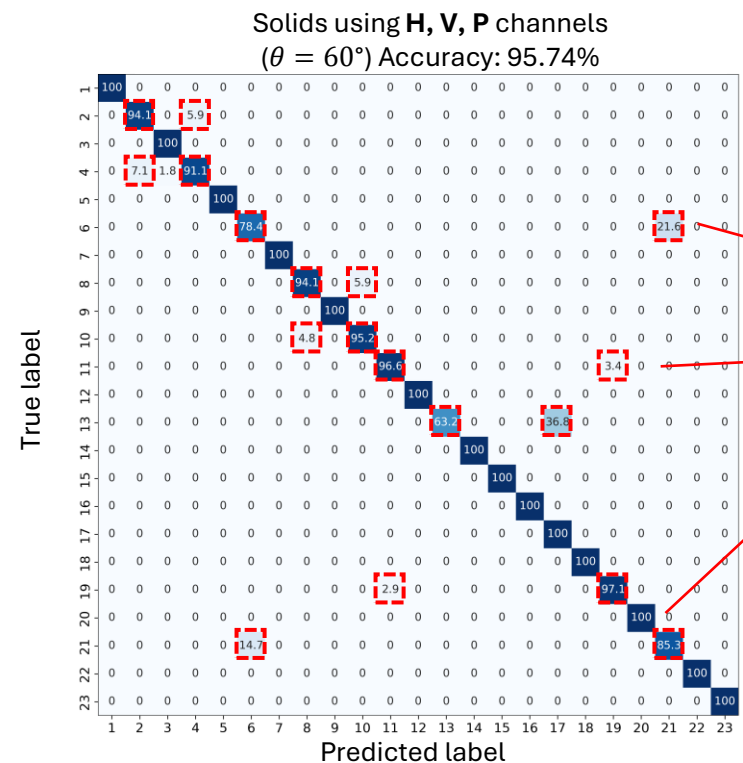
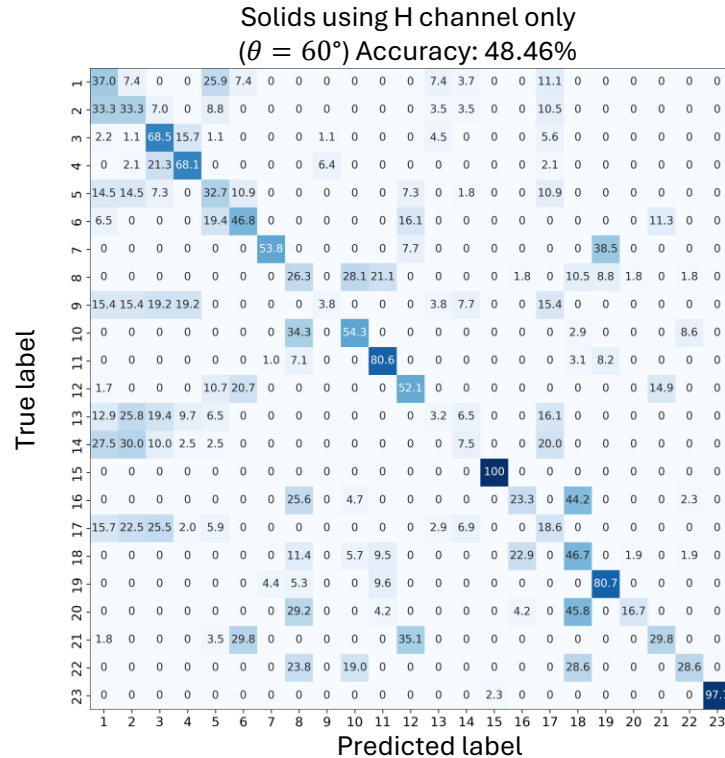
Predicted label

w/ PolySight

Predicted label

w/o/ PolySight

w/ PolySight



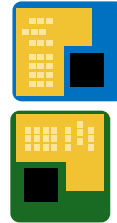
Materials	ϵ'	$-\text{j}\epsilon''$
Solids (1–19)		
1. Metal	1.29e5	3.05e5
2. Acrylic (1 mm)	3.56	0.18
3. Acrylic (3 mm)	4.00	0.40
4. Polycarbonate (1 mm)	4.26	0.21
5. PET (0.5 mm)	3.78	0.70
6. PET (1 mm)	3.83	0.73
7. Vinyl (0.37 mm)	3.65	0.70
8. PVC	4.15	0.81
9. PLA (ender, 1 mm)	3.79	0.34
10. PLA (hyper, 1 mm)	4.21	0.30
11. TPU (1 mm)	3.91	0.21
12. Ceramic	4.75	2.43
13. Rubber	5.06	1.11
14. Silicone Rubber	3.52	0.82
15. Glass (2 mm)	7.26	0.85
16. Glass Fiber	4.69	0.70
17. Paper (smooth)	4.63	0.31
18. Cardboard	5.24	0.01
19. Cardboard (rough)	4.54	0.56

Materials	ϵ'	$-\text{j}\epsilon''$
Solids (20–23)		
20. Fabric	2.21	0.13
21. Wood (thin, smooth)	2.61	0.71
22. Wood (rough)	5.02	0.81
23. Cork	10.6	N/A
Liquids		
24. Soy Sauce (w/ glass)	4.30	0.53
25. Corn Oil (w/ glass)	6.01	2.69
26. Corn Syrup (w/ glass)	6.42	0.05
27. Water (w/ glass)	3.66	0.43
28. Cane Sugar (1.3%, w/ glass)	3.72	0.43
29. Cane Sugar (3.8%, w/ glass)	3.78	0.44
30. Cane Sugar (5.1%, w/ glass)	3.81	0.43
31. Cane Sugar (6.3%, w/ glass)	3.89	0.41
32. Cane Sugar (7.4%, w/ glass)	3.96	0.39
Powders		
33. Sugar Powder (w/ plastic)	4.46	0.42
34. Salt (w/ plastic)	4.30	0.94
35. Protein Powder (w/ plastic)	5.74	0.45
36. Milk Powder (w/ plastic)	4.88	0.74

One more thing about the distributed setup



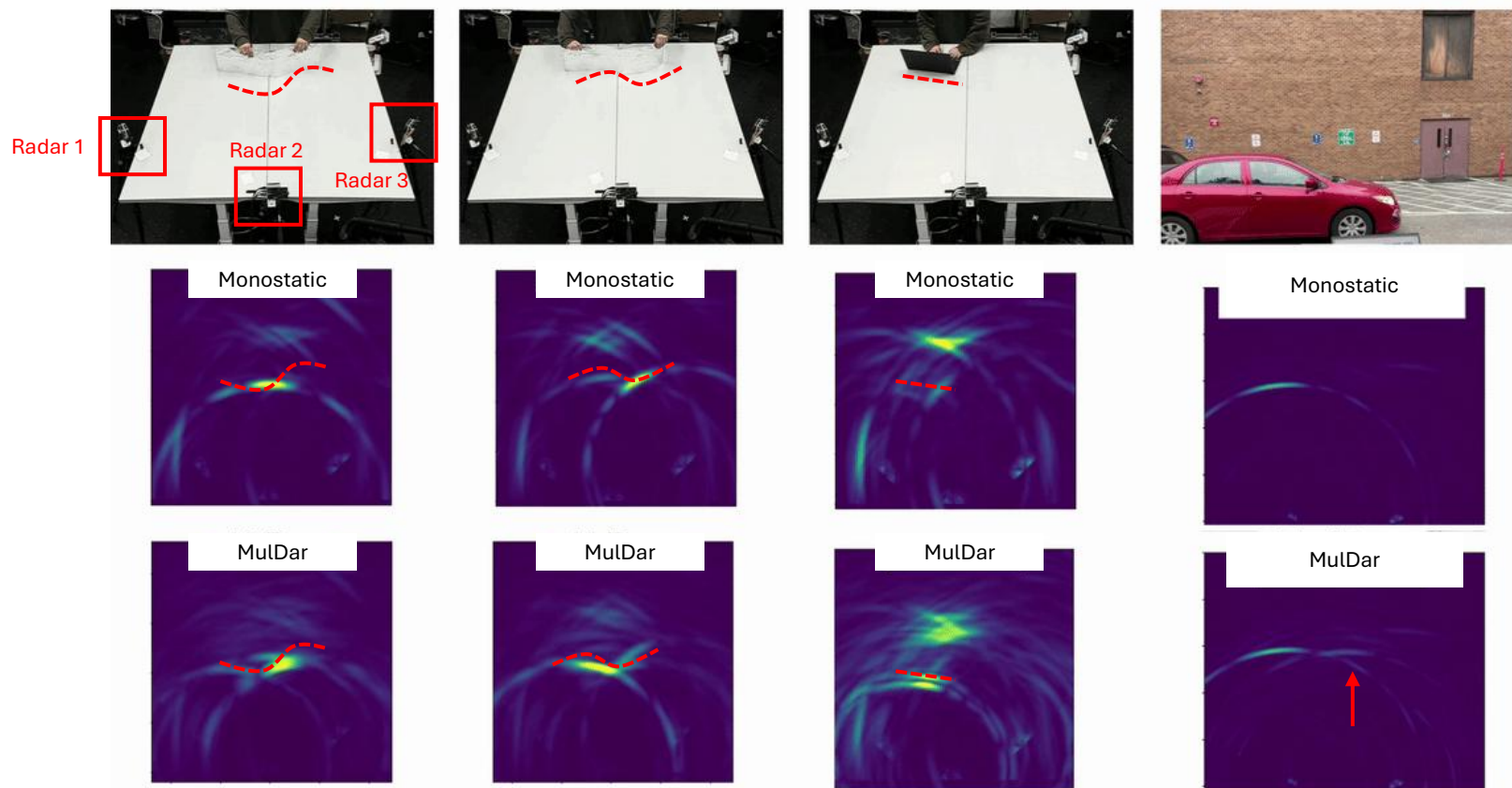
One more thing about the distributed setup



One more thing about the distributed setup



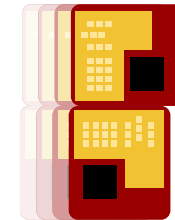
One more thing about the distributed setup



MulDar: Unleashing the Potential of Distributed COTS mmWave Radar by Exploiting Cross-Device Channels
Sensys Demo, Mobisys 2026



Towards Practical Bi-Static Polarimetric Imaging Using Commodity mmWave Radars for Material Sensing



PolySight project page



More about the
distributed setup



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